PRASANTH AYITAPU

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Data Scientist at IIM Visakhapatnam's IDeAL Research Lab building and deploying analytics, machine learning and decision-support applications on public-sector and research data. Experience spans data extraction from unstructured sources, predictive modeling, NLP, computer vision and Linux-based application deployment. Co-author research papers across machine learning, cybersecurity, IoT and astronomy.

SKILLS

Languages: Python, SQL, C, Bash, HTML/CSS

Data & Backend: Pandas, NumPy, PostgreSQL, SQLite, FastAPI, Flask, ETL Pipeline Design, Data Validation, pdfplumber, tabula

AI/ML: Scikit-learn, XGBoost, CatBoost, LightGBM, TensorFlow, Keras, SHAP, GANs, LSTM, BERTopic, Time-Series Forecasting, OpenCV

Visualization: Plotly, Matplotlib, Seaborn, Streamlit, Power BI, Excel

Tools & Platforms: Git/GitHub, Linux (Ubuntu), Gunicorn, Jupyter, VS Code, WordPress, Google Colab

WORK EXPERIENCE

Data Scientist | IDeAL Research Lab, IIM Visakhapatnam **Jul 2025 – Present**

Visakhapatnam, India

AP Groundwater Sustainability Dashboard [Live]

- **Developed a Flask-based groundwater analytics application** covering 13 indicators across all 26 Andhra Pradesh districts using CGWB data from 2022–2025, forecasting 2026 values with ensemble regression models to support district-level resource planning.
- Engineered an explainable risk-analysis workflow using Random Forest and SHAP, mapping model outputs to CGWB extraction categories and generating district-level visualizations and rule-based insights for interpreting groundwater conditions.
- Deployed on Linux with Gunicorn, SSL/HTTPS and a UFW firewall; featured in **The Times of India** as a public-policy data product from IIM Visakhapatnam. Aligned with UN SDGs 6 and 13.
- ***Keywords:*** *Python, Flask, Scikit-learn, SHAP, Pandas, Plotly, Linux, Gunicorn*

NIRF Decision Support System [Live]

- Designed a Flask-based decision-support application for NIRF Management rankings using 401 institute-year records and 15 ranking sub-scores, incorporating interactive Plotly analytics with 2026 score and rank forecasting.
- Benchmarked various model configurations using chronological walk-forward validation, comparing predictive accuracy and ranking quality against a previous-score baseline.
- Chose a simple, explainable score-transition Linear Regression for production; added validation for duplicate institute-year records and CSV/Excel ingestion and confirmed the five-category NIRF weighting reconstructs the recorded score to within 0.01 mean absolute error.
- ***Keywords:*** *Python, Flask, Scikit-learn, Pandas, Plotly, Time-Series Validation, Gunicorn*

Road Accident Intelligence Dashboard

- Engineered automated Python pipelines to extract and structure multi-year accident data from official MoRTH PDF reports into a unified time-series dataset covering 50 major Indian cities.
- Trained and benchmarked three GAN architectures (TimeGAN, CTGAN, WGAN-GP) to generate synthetic estimates for missing-year records, enabling a complete longitudinal analysis of an otherwise incomplete government dataset.
- Built a Plotly-powered Flask dashboard with auto-generated narrative insights across time-series, cross-city and summary modes, plus a composite safety index, K-Means city clustering, India map view, SHAP explanations and LSTM forecasting. Research paper submitted for peer review.
- ***Keywords:*** *Python, Flask, Pandas, Plotly, GANs, TensorFlow, SHAP, pdfplumber, tabula*

Polavaram Project NLP Corpus & Stakeholder Analytics

- **Established a resumable web-scraping and data-ingestion pipeline** using SQLite-based URL queuing, sitemap/RSS discovery, rate limiting and multi-layer deduplication, producing a 1,626-document corpus containing 6.5M+ English words.
- Applied a stakeholder analytics framework combining a 20-stakeholder taxonomy, rule-based sentiment and impact classification, and BERTopic with MiniLM embeddings, UMAP and HDBSCAN to identify major discourse themes.
- Delivered an 8-tab Streamlit analytics dashboard and local BM25 retrieval system using SQLite FTS5, enabling source-cited document search and question answering without requiring an external LLM.
- ***Keywords:*** *Python, Web Scraping, NLP, BERTopic, SQLite, Streamlit, Plotly, Pandas*

ICGE 2026 Conclave - Digital Communications & Event Operations [Link]

- Developed the official ICGE 2026 conference website on WordPress and managed its content structure and live updates through the event.
- Handled communication for 200+ submitted participants via Outlook mail merge; maintained Excel records of registrations, payments and selections; coordinated queries with faculty and external professors, and logistics (accommodation, refreshments, rooms) with campus IT and staff.
- Designed event posters and coordinated consistent messaging with the IIM-V social media team.
- ***Keywords:*** *WordPress, Canva, Excel, Mail Merge, Cross-functional Coordination*

PROJECTS

E-Commerce Risk & Analytics API 2026

- Designed a PostgreSQL analytics schema across 9 related tables containing 99,441 orders, using indexes, foreign keys and relational constraints to support revenue, seller, delivery, customer and risk analysis.
- Implemented 11 documented FastAPI endpoints with Pydantic validation and SQL queries using CTEs, window functions, NTILE and percentile-based analysis to expose business analytics through a REST API.
- Performed RFM customer segmentation and rule-based risk analysis across 94,687 customers, identifying purchasing patterns and tuning risk thresholds based on observed order-value and delivery behavior.
- ***Keywords:*** *PostgreSQL, FastAPI, Python, SQL, Pydantic, psycopg2, Uvicorn*

Customer Behavior Analysis

- Built an end-to-end analytics workflow on customer shopping data: cleaning, validation and EDA in Python (Pandas, Matplotlib, Seaborn), followed by SQL business analysis in a relational database.
- Wrote SQL queries on revenue by segment and age group, subscriber vs non-subscriber spend, discount-dependent products, repeat customers and shipping type; built an interactive Power BI dashboard and a written business report with recommendations.
- ***Keywords:*** *Python, Pandas, SQL, Power BI, Matplotlib, Seaborn, Git*

IPL Score Prediction System Jun 2023 – Dec 2023

- Built a score-prediction pipeline on 76,015 IPL records (2008–2017) sourced from ESPN and Cricbuzz, using one-hot encoded batting and bowling teams with runs, wickets, overs and last-5-over runs and wickets as features, and a chronological train/test split.
- Compared Linear Regression (R^2 = 0.75), SVM (0.73) and Random Forest (0.71); deployed the models in a Flask web interface that returns a ±5 run range to communicate prediction uncertainty. Co-authored the research paper.
- ***Keywords:*** *Python, Flask, Scikit-learn, Pandas, SVM, Random Forest, Feature Engineering*

Image Enhancement using Residual U-Net and PatchGAN 2024

- Engineered a GAN-based image enhancement pipeline combining a residual U-Net generator, PatchGAN discriminator and VGG16 perceptual loss, trained on DIV2K with adversarial, pixel-wise and perceptual objectives.
- Achieved average PSNR of 27.64 and SSIM of 0.9270 on the validation set, evaluating enhancement quality across photographs, paintings, underwater imagery, medical images and driving scenes. Co-authored IEEE paper.
- ***Keywords:*** *Python, TensorFlow, Keras, GANs, VGG16, OpenCV, Computer Vision*

Photometric Prediction of Star Formation Rates (27M+ Galaxies) Jan 2024 – Jun 2024

- Processed 27M+ SDSS-DR7 galaxy records using quality filtering, redshift and magnitude constraints, cross-matching and normalization to prepare photometric data for specific star formation rate prediction.
- Benchmarked Linear Regression, SVR, Random Forest, Decision Tree, Gradient Boosting, LSTM and neural-network models using cross-validation to evaluate photometry-based prediction approaches.
- Developed a photometry-based prediction framework for large-scale sSFR estimation, demonstrating both the potential and limitations of using photometric measurements as an alternative to direct spectroscopic analysis.
- ***Keywords:*** *Python, Scikit-learn, TensorFlow, LSTM, Pandas, Feature Engineering*

Malware Detection Across Evolving Threat Landscapes 2025

- Benchmarked XGBoost, CatBoost, LightGBM, Random Forest, CNN and RNN-LSTM models across EMBER2018 and CIC-Evasive-PDFMal2022 to evaluate malware detection under conventional and evasive threat scenarios.
- Evaluated model robustness using FGSM adversarial perturbations, comparing tree-based, deep-learning and ensemble approaches across legacy and evasive malware datasets.
- ***Keywords:*** *Python, XGBoost, CatBoost, CNN, RNN-LSTM, Ensemble Methods, Cybersecurity*

Automated Parking System Sep 2022 – Mar 2023

- Designed an automated parking allocation system using Arduino UNO and ultrasonic sensors, detecting vehicle presence and assigning available parking slots with duplicate-occupancy checks.
- Implemented a Python/OpenCV number-plate recognition pipeline using grayscale conversion, noise reduction, edge detection, contour-based plate localization and OCR, evaluated on a 20,000+ image dataset.
- ***Keywords:*** *Python, Arduino, OpenCV, OCR, IoT, Embedded Systems*

PUBLICATIONS (CO-AUTHOR)

- An Efficient Automatic Car Parking Slot Allocation System — ICSC 2025, JIIT New Delhi
- Data-Driven Cricket: A Machine Learning Approach to IPL Score Prognostication
- Enhancing Perceptual Quality of Images using Deep Residual U-Net and PatchGAN Discriminator — IEEE
- Bridging Legacy and Modern Threat Detection: An Analysis of Machine Learning Models on EMBER2018 and CIC-EvasivePDF2022

EDUCATION

Amrita Vishwa Vidyapeetham, Bengaluru Sep 2021 – Aug 2025

B.Tech, Electronics and Computer Engineering | **CGPA: 7.05 / 10**

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Image Processing, Pattern Recognition, DBMS, Computer Networks, Big Data Analytics, Cybersecurity, Operating Systems, Project Management

National Centre for Excellence, Bengaluru 2021

CBSE Class XII

CERTIFICATIONS

McKinsey Forward Program – McKinsey & Company Feb 2026

100 Days of Code: The Complete Python Pro Bootcamp – Udemy Sep 2023

Supervised Machine Learning: Regression and Classification – Coursera Apr 2023

ACHIEVEMENTS & ACTIVITIES

- **AP Groundwater Dashboard** featured in **The Times of India** as a data-driven public-policy initiative from IIM Visakhapatnam.
- Presented “An Efficient Automatic Car Parking Slot Allocation System” at **IEEE SPCOM 2025 (ICSC-2025)**, JIIT New Delhi.
- **Executive Member, Vyom Club & Sankya Club**, Amrita Vishwa Vidyapeetham (Jan 2022 – Jun 2023): coordinated technical and cultural events, scheduling, logistics and participant communication.